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THE NATIONAL COLLEGE BASAVANAGUDI, BENGALURU- 560 004

AUTONOMOUS NAAC Accredited 'A' Grade

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Arvind N from National College, Jayanagar, conducted an insightful Python introduction session on 27 September 2024 for first-year mathematics students from National College, Basavanagudi. This session was designed to introduce students to the basics of Python programming with a focus on how the language can be applied in mathematical problem-solving and computational tasks. Recognizing the increasing importance of Python in fields like data analysis, scientific computing, and mathematical modeling, Arvind emphasized the relevance of Python to the students' future academic and professional pursuits.

The session began with an introduction to Jupyter Notebook, an interactive platform used for writing and executing Python code. Arvind demonstrated how Jupyter's interface allows students to write code in small, manageable sections called cells, which can be executed individually. This format is particularly beneficial for solving mathematical problems as it enables step-by-step code execution and real-time analysis of

results. Students were guided through the setup of Jupyter Notebook, and they quickly became familiar with using it to write and run Python scripts.

In terms of content, the session covered essential arithmetic operations, where students were introduced to Python's syntax for performing basic calculations such as addition, subtraction, multiplication, and division. Arvind made these operations relatable by linking them to mathematical concepts that students encounter in their coursework. This provided them with a practical understanding of how Python can simplify numerical calculations.

Next, Arvind introduced conditional statements (if, else), a core programming concept that allows for decision-making in code. These statements are fundamental for writing algorithms, and Arvind provided examples where students could use conditional logic to solve mathematical problems. The students practiced writing small programs where conditions were used to control the flow of calculations, helping them grasp how programming logic integrates with mathematical operations.

The session also included a segment on trigonometric functions, a vital part of the mathematics curriculum. Using Python's built-in math library, students learned how to calculate trigonometric values such as sine, cosine, and tangent. Arvind demonstrated how Python can be used to compute and visualize trigonometric functions, showing its effectiveness as a tool for handling more advanced mathematical problems.

Throughout the session, students engaged in hands-on practice, using Jupyter Notebook to perform arithmetic, apply conditional statements, and compute trigonometric functions. This practical approach allowed them to immediately apply what they were learning, reinforcing their understanding of both Python and its application to mathematics. Arvind's ability to connect Python programming to the students' academic needs made the session highly relevant and engaging.

In conclusion, the session provided the students of National College, Basavanagudi, with a solid introduction to Python programming. They left with a strong grasp of using Python for basic mathematical computations and a new tool in their academic toolkit. Arvind N's clear instruction and emphasis on practical application ensured that the students were well-

prepared to continue exploring Python's potential in their mathematical studies. This session not only sparked interest in Python but also set the stage for further exploration of its capabilities in advanced mathematical problem-solving and computational tasks.



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